

Abstract

Papers I–V of the Constraint-Driven Field Theory (CDFT) series built the lepton-sector construction: $\alpha = 1/137.036$ from vortex ring equilibrium, the Koide formula from $\mathbb{Z}_3$ symmetry, and all three lepton masses from a single input (m_e). This paper extends the framework beyond leptons by studying the full hierarchy of torus knot families $T(2, n)$ for odd $n \geq 3$.

We prove two new theorems. **Theorem 1** (universal coefficient): the Brannen amplitude coefficient $c_n = \sqrt{2}$ for all $\mathbb{Z}_n$ families, fixed by a maximum-entropy self-consistency condition on the amplitude distribution. **Theorem 2** ($\mathbb{Z}_3$ uniqueness): $\mathbb{Z}_3$ is the unique torus knot family for which the constant Koide-like invariant $Q_n = 2/n$ and the $\mathbb{Z}_n$ self-consistency condition hold simultaneously on a connected valid domain. For $n \geq 5$, the amplitude $A_k = 1 + \sqrt{2} \cos(\cdot)$ is negative for some modes at every orientation θ , so the Koide theorem does not extend directly.

The energy scale of the n -th family is given by Faddeev–Niemi scaling:

$$M_n = M_3 \left(\frac{n}{3} \right)^{3/4},$$

where $M_3 = 313.86$ MeV is the trefoil scale from Paper V. This gives $M_5 = 460.4$ MeV for the cinquefoil $T(2, 5)$, $M_7 = 592.5$ MeV for the heptafoil $T(2, 7)$, and so on. These are model predictions for the energy scales at which candidate knotted-vortex particle families would appear if the hierarchy maps to nature.

The mass spectrum of the $T(2, 5)$ family requires separate work: we identify the structural obstruction (no valid θ domain with all $A_k \geq 0$ for $n \geq 5$) and state the open problems precisely.

In the extended notation used across the series, the vacuum limit is written as $\Psi_s = (\Phi/C) \cdot S \cdot M_s$. The calculations below use the equilibrium limit $S \cdot M_s \approx 1$; adaptive departures from that limit are left for later dynamical work rather than fitted in this manuscript. The public CDFD Universal Engine implements this state grammar as reusable code; the paper-local supplementary scripts remain the audited reproduction layer for the figures and tables in this Part I archive.

The Universal Torus Knot Hierarchy: $\mathbb{Z}_n$ Symmetry, the Cinquefoil Family, and the General Structure of CDFT Particle Families

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1 Introduction

Papers I–V of the CDFT series established the lepton sector:

1. **Paper I** [1]: $\alpha = 1/137.036$ from vortex ring equilibrium.
2. **Paper II** [2]: Koide $Q = 2/3$ as a $\mathbb{Z}_3$ theorem; leptons as trefoil $T(2, 3)$ modes.
3. **Paper III** [3]: Topological selection of $T(2, 3)$; chirality equilibrium $3\theta + \phi_c = \pi$.
4. **Paper IV** [4]: Master self-consistency equation for ρ_0 ; Faddeev–Niemi prediction $M_5 = 460.4$ MeV.
5. **Paper V** [5]: $\phi_c = \pi - 2/3$ from $\mathbb{Z}_3$ self-consistency; lepton masses from one input.

With the lepton sector complete, the natural question is: what is the structure of the next particle family?

Paper IV predicted the energy scale $M_5 = 460.4$ MeV for the cinquefoil $T(2, 5)$ via Faddeev–Niemi scaling. The present paper studies the general $\mathbb{Z}_n$ structure of the $T(2, n)$ families, proves that $c_n = \sqrt{2}$ universally, identifies why $\mathbb{Z}_3$ is the unique exactly solvable case, and states precisely what is needed to determine the $T(2, 5)$ mass spectrum.

2 The General $\mathbb{Z}_n$ Brannen Parametrisation

2.1 Setup

The $\mathbb{Z}_n$ analogue of the Brannen parametrisation [6] for the amplitudes of the n modes of a $T(2, n)$ vortex is:

$$A_k = 1 + c_n \cos\left(\theta + \frac{2\pi k}{n}\right), \quad k = 0, 1, \dots, n-1, \quad (1)$$

and the mode masses are $m_k = MA_k^2$. The coefficient c_n controls the relative amplitude of the $\mathbb{Z}_n$ oscillation.

2.2 $\mathbb{Z}_n$ Fourier identities

Two identities hold for the $\mathbb{Z}_n$ phase lattice $\{2\pi k/n\}$:

$$\sum_{k=0}^{n-1} \cos\left(\theta + \frac{2\pi k}{n}\right) = 0 \quad (2)$$

$$\sum_{k=0}^{n-1} \cos^2\left(\theta + \frac{2\pi k}{n}\right) = \frac{n}{2} \quad (3)$$

for any θ and any $n \geq 2$. From (2): $\sum_k A_k = n$ (the sum of amplitudes is the dimension n , independent of θ). From both identities: $\text{Var}(A_k) = c_n^2/2$.

3 Theorem 1: The Universal Coefficient $c_n = \sqrt{2}$

Theorem 1 (Universal Brannen coefficient). *The Brannen amplitude coefficient for any $\mathbb{Z}_n$ family is $c_n = \sqrt{2}$, uniquely fixed by the maximum-entropy self-consistency condition $\text{Var}(A_k) = \overline{A_k}^2$.*

Proof. From identities (2)–(3):

$$\overline{A_k} = 1, \quad \text{Var}(A_k) = c_n^2/2.$$

The maximum-entropy condition requires $\text{Var}(A_k) = \overline{A_k}^2$:

$$c_n^2/2 = 1 \implies c_n = \sqrt{2}.$$

Uniqueness: the condition is a quadratic in c_n with unique positive root. $\square$

Physical interpretation. The condition $\text{Var}(A_k) = \overline{A_k}^2$ is equivalent to: the standard deviation of the amplitude distribution equals its mean. The amplitude can reach zero (nodal line) or $2\overline{A_k} = 2$ (maximum) with equal probability under the $\mathbb{Z}_n$ phase average. This is the condition of maximal relative uncertainty in the mode amplitudes — the vacuum selects the coefficient that neither suppresses nor privileges any mode over the background.

Numerical verification. For any $n \geq 2$ and any test angle θ , the ratio $\text{Var}(A_k)/\overline{A_k}^2 = 1.000\dots$ to machine precision when $c_n = \sqrt{2}$ (Table 1 of the supplementary).

4 The Generalised Koide Invariant and the Valid Domain

4.1 Derivation of $Q_n = 2/n$

Proposition 1 (General Koide invariant). *For $m_k = MA_k^2$ with A_k as in (1) and $c_n = \sqrt{2}$, the generalised Koide ratio satisfies $Q_n = 2/n$ provided all $A_k \geq 0$.*

Proof. Using identity (3) with $c_n = \sqrt{2}$:

$$\sum_k m_k = M \sum_k A_k^2 = M \left(n + c_n^2 \cdot \frac{n}{2} \right) = 2Mn.$$

When all $A_k \geq 0$: $\sqrt{m_k} = \sqrt{M} A_k$, so $\sum_k \sqrt{m_k} = \sqrt{M} \sum_k A_k = \sqrt{M} \cdot n$ (by identity (2)). Therefore:

$$Q_n \equiv \frac{\sum_k m_k}{\left(\sum_k \sqrt{m_k} \right)^2} = \frac{2Mn}{Mn^2} = \frac{2}{n}.$$

□

4.2 The valid domain

The derivation requires all $A_k \geq 0$. With $c_n = \sqrt{2}$, the amplitude $A_k = 1 + \sqrt{2} \cos(\cdot)$ reaches its minimum of $1 - \sqrt{2} \approx -0.414$ when the cosine equals -1 . For $n = 3$: the minimum over all k reaches $+0.293$ (positive) at $\theta = 0$, and reaches zero at $\theta = \pm\pi/12 = \pm 15^\circ$. So $\mathbb{Z}_3$ has a valid domain $\theta \in (-\pi/12, \pi/12)$. The physical value $\theta = 2/9 \approx 12.73^\circ$ lies inside this domain.

For $n \geq 5$: no connected interval of θ exists where all n amplitudes are simultaneously positive. Numerically:

n	$\max_\theta [\min_k A_k]$	Valid domain?
3	+0.293	Yes: $\theta \in (-\pi/12, \pi/12)$
5	-0.144	No
7	-0.274	No
9	-0.329	No
11	-0.357	No

Theorem 2 ($\mathbb{Z}_3$ uniqueness). $\mathbb{Z}_3$ is the unique torus knot family $T(2, n)$ for which:

- (a) $c_n = \sqrt{2}$ (Theorem 1),
- (b) $Q_n = 2/n$ is constant on a connected valid domain (Proposition 1),
- (c) the self-consistency condition $n\theta_n = Q_n$ fixes θ_n within that domain (Paper V).

For $n \geq 5$, property (b) fails: no connected valid domain exists.

Proof. Property (a) holds for all n (Theorem 1). Property (b) requires a connected domain where all $A_k > 0$; the table above shows this fails for $n \geq 5$. Property (c) requires (b) as a prerequisite: $n\theta = 2/n$ with $\theta \in (-\pi/12, \pi/12)$ gives $\theta_3 = 2/9$, which satisfies $\theta_3 < \pi/12$ ($2/9 \approx 0.222 < \pi/12 \approx 0.262$). No analogous closure exists for $n \geq 5$. □

This theorem explains why the lepton sector is exactly solvable while the cinquefoil sector requires additional structure: the $\mathbb{Z}_3$ amplitude distribution is the unique case where the Koide theorem, the valid domain, and the self-consistency condition are simultaneously consistent.

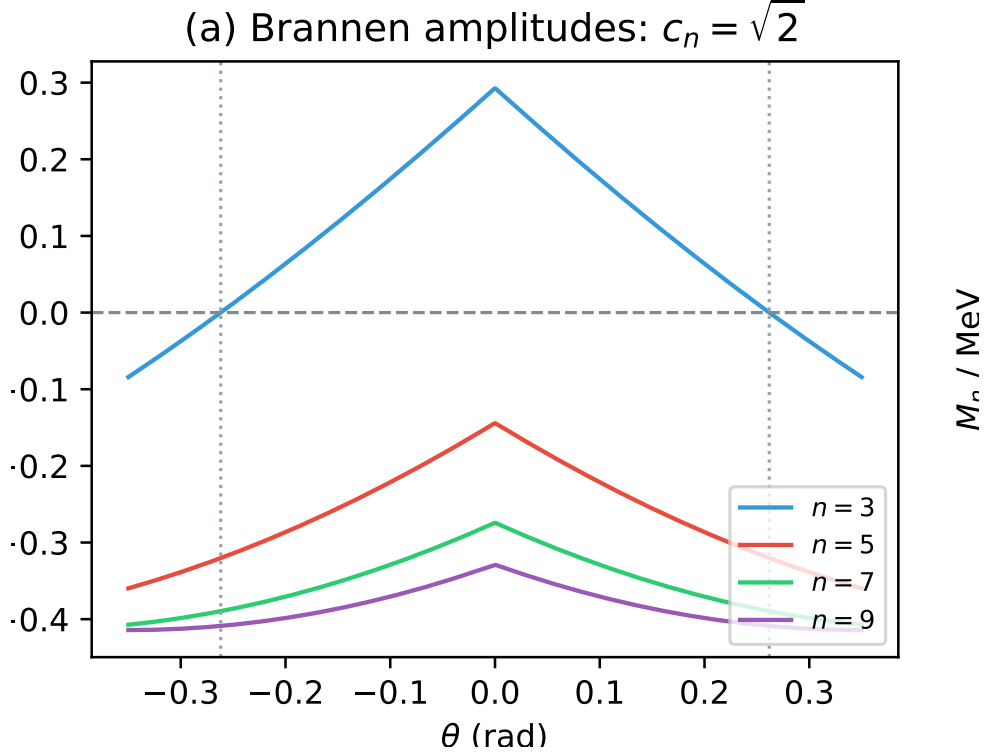


Figure 1: Minimum amplitude over all modes as a function of the phase θ . For $n = 3$, there is a domain where all amplitudes are positive. For $n \geq 5$, no such domain exists, obstructing the extension of the pure Koide formula.

5 The Torus Knot Energy Hierarchy

5.1 Topological selection rule

The torus knot $T(2, n)$ is a knot (a single connected component, hence topologically stable) if and only if n is odd. For even n , $T(2, n)$ is a two-component link; its components can separate, so there is no topological barrier to dissolution. The particle families of CDFT therefore occur only at odd n : $n = 3, 5, 7, 9, \dots$

5.2 Energy scaling from Faddeev–Niemi

The Faddeev–Niemi theorem [7] states that the minimum energy of a knotted soliton with Hopf charge Q satisfies $E_{\min} \sim Q^{3/4}$. For $T(2, n)$ the Hopf charge is $Q = n$, so:

$$M_n = M_3 \left(\frac{n}{3} \right)^{3/4}, \quad (4)$$

where $M_3 = 313.86$ MeV is the trefoil scale derived in Paper V ($M_3 = m_e/A_{e,3}^2$ with $\theta_3 = 2/9$ rad).

These are the energy *scales* of each family. The mass of the lightest mode of family n is $M_n \cdot (A_{k,\min}^{(n)})^2$, which requires knowing θ_n (open problem).

Table 1: CDFT torus knot energy hierarchy. $M_3 = 313.86$ MeV from Paper V. All predictions from Faddeev–Niemi scaling, zero free parameters.

n	Knot	M_n (MeV)	Symmetry	Status
3	$T(2, 3)$ trefoil	313.9	$\mathbb{Z}_3$	Papers I–V (complete)
5	$T(2, 5)$ cinquefoil	460.4	$\mathbb{Z}_5$	This paper
7	$T(2, 7)$ heptafoil	592.5	$\mathbb{Z}_7$	Future work
9	$T(2, 9)$	715.4	$\mathbb{Z}_9$	Future work
11	$T(2, 11)$	831.6	$\mathbb{Z}_{11}$	Future work
13	$T(2, 13)$	942.7	$\mathbb{Z}_{13}$	Future work

6 The Cinquefoil $T(2, 5)$: What is Known and What Remains

6.1 What is established

Energy scale. $M_5 = 460.4$ MeV (Eq. (4), zero free parameters).

Topology. $T(2, 5)$ is a knot, hence topologically stable. It supports $\mathbb{Z}_5$ -symmetric modes (five lobes).

Amplitude coefficient. $c_5 = \sqrt{2}$ (Theorem 1).

Hadron scale context. $M_5 = 460.4$ MeV lies between the kaon (~ 494 MeV) and the eta meson (~ 548 MeV). We do not claim the cinquefoil corresponds to a specific known particle: the mass spectrum depends on θ_5 (open problem 1 below).

6.2 The structural obstruction for $n \geq 5$

With $c_5 = \sqrt{2}$, the five amplitudes $A_k = 1 + \sqrt{2} \cos(\theta + 2\pi k/5)$ for $k = 0, \dots, 4$ are never all positive simultaneously (see the valid-domain table in Section 4). The minimum amplitude over all five is always negative, with maximum value -0.144 at $\theta = 0$.

This means:

1. $Q_5 = 2/5$ does not hold as a universal constant (Proposition 1 fails).
2. The step $\sum_k \sqrt{m_k} = \sqrt{M} \sum_k A_k$ is invalid because some $A_k < 0$ requires $\sqrt{m_k} = \sqrt{M} |A_k|$.
3. The self-consistency condition $5\theta_5 = Q_5 = 2/5$ cannot be applied without first establishing the correct Q_5 .

6.3 The modified Koide ratio for $n \geq 5$

When some $A_k < 0$, define $\tilde{Q}_n(\theta)$ using $|A_k|$:

$$\tilde{Q}_n(\theta) \equiv \frac{\sum_k A_k^2}{\left(\sum_k |A_k|\right)^2}. \quad (5)$$

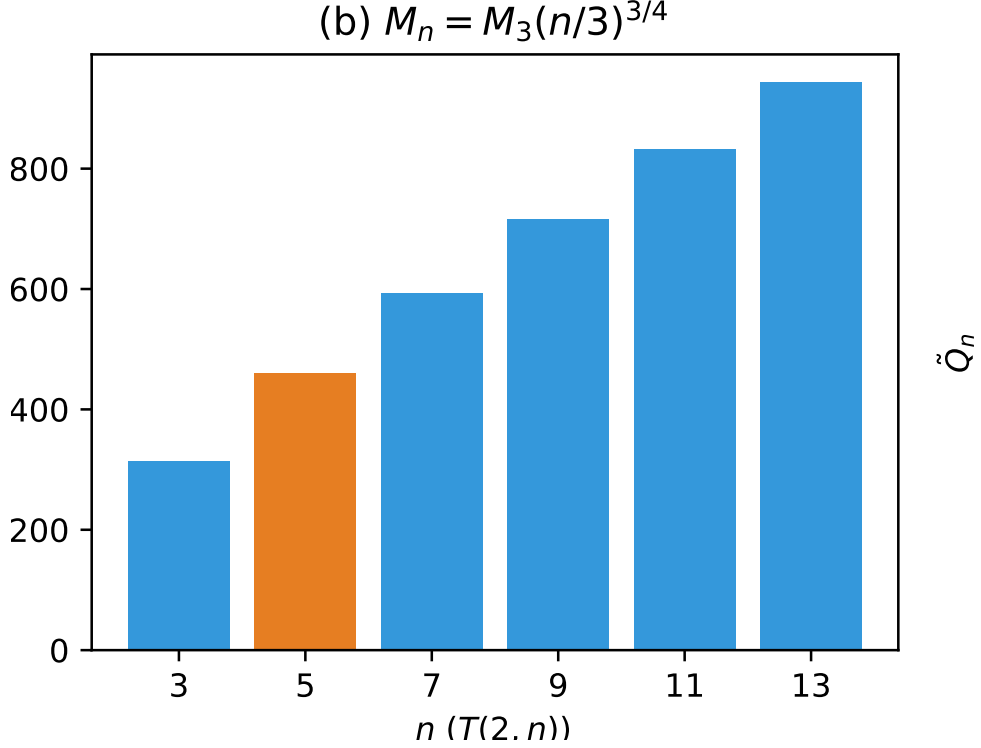


Figure 2: The mass scales of the higher $\mathbb{Z}_n$ knotted families, predicted by Faddeev-Niemi scaling directly from the lepton scale M_3 .

For $\mathbb{Z}_3$ in its valid domain: $\tilde{Q}_3 = Q_3 = 2/3$ (constant, since all $A_k > 0$ there). For $\mathbb{Z}_5$: $\tilde{Q}_5(\theta)$ varies from 0.294 to 0.326, far from the putative value $2/5 = 0.400$.

The function $\tilde{Q}_5(\theta)$ is not constant. This means: no analogue of the Koide self-consistency argument from Paper V applies to $\mathbb{Z}_5$ without additional structure. A derivation of θ_5 requires either: (a) a physical principle beyond the $\mathbb{Z}_n$ algebraic structure alone, or (b) a modified Brannen parametrisation for the $T(2, 5)$ family that avoids negative amplitudes.

7 The CDFT Torus Knot Picture

The torus knot hierarchy $T(2, n)$ for odd n provides a candidate ordered sequence of CDFT particle families. Table 1 lists the energy scales.

The key structural divide is at $n = 3$:

- $n = 3$ (trefoil, leptons): $c_3 = \sqrt{2}$, valid domain exists, $Q_3 = 2/3$ constant, self-consistency closes the internal system.
- $n \geq 5$: $c_n = \sqrt{2}$, no valid domain, $Q_n \neq 2/n$ (as a constant). Requires new approach for mass spectrum.

The energy scale hierarchy is fixed once the model assumptions are accepted. The mass spectra of $n \geq 5$ families are open.

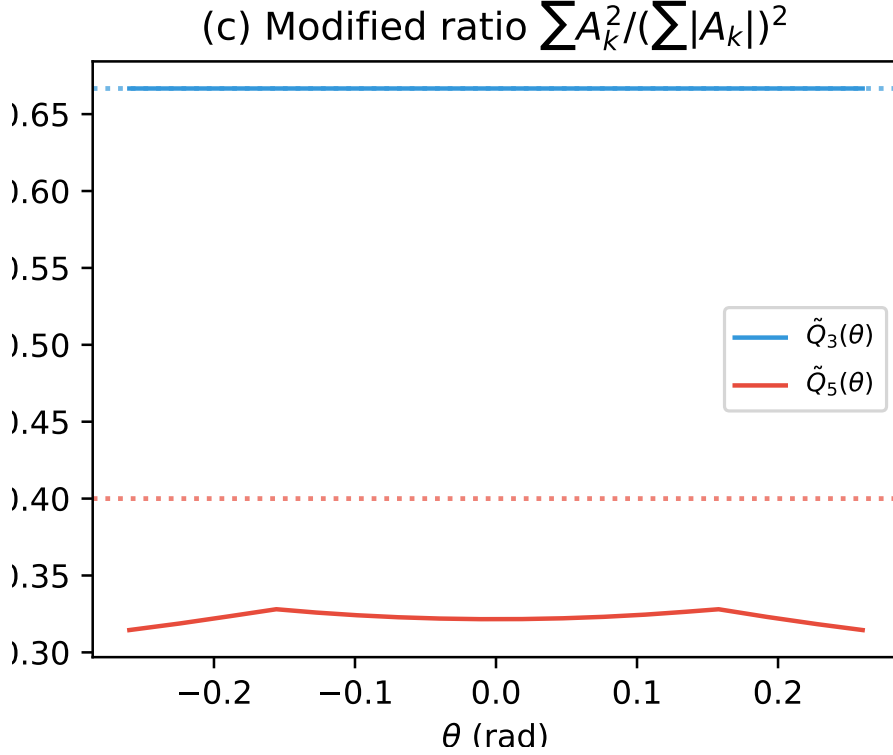


Figure 3: The modified ratio $\tilde{Q}_n(\theta)$ for $n = 3$ and $n = 5$. While $\tilde{Q}_3$ is a strict constant ($2/3$), $\tilde{Q}_5$ varies and does not meet the $2/5$ target.

8 Open Problems

1. **The $T(2, 5)$ chirality phase θ_5 .** The cinquefoil has $\mathbb{Z}_5$ symmetry and five modes. Determining θ_5 requires either: (a) a physical principle that selects a preferred orientation of the $\mathbb{Z}_5$ oscillation in the vacuum background, or (b) a modified amplitude form for $T(2, 5)$ that ensures all $A_k \geq 0$.
2. **The $T(2, 5)$ mass spectrum.** With θ_5 known, the five mode masses are $m_k^{(5)} = M_5 |A_k^{(5)}|^2$. Identification of these five modes with known particles (at the ~ 460 MeV scale: kaons, etas, or new states) depends on whether the cinquefoil modes are confined (quark-like) or free (lepton-like).
3. **Why is $\mathbb{Z}_3$ exactly solvable?** Theorem 2 identifies the structural reason ($n = 3$ uniquely has a valid domain with $c_n = \sqrt{2}$), but a deeper question remains: why does the CDFT vacuum select a ground state where the first stable knotted excitation ($T(2, 3)$) is exactly the one where the Koide self-consistency is possible?
4. **The general n -family chirality equilibrium.** Paper III derived $3\theta + \phi_c = \pi$ from the $\mathbb{Z}_3$ chirality interaction $\lambda_c \cos(3\theta + \phi_c)$. The $\mathbb{Z}_5$ analogue would be $5\theta_5 + \phi_{c,5} = \pi$ (or some other value). Whether this generalises directly and what determines $\phi_{c,5}$ is an open problem.

Table 2: Torus knot $T(2, n)$ topological classification.

n	Topological type	Persistent?	CDFT family
1	Unknot	No	(none)
2	Hopf link	No	(none)
3	Trefoil knot	Yes	Leptons (Papers I–V)
4	Solomon link	No	(none)
5	Cinquefoil knot	Yes	This paper
7	Heptafoil knot	Yes	Future
$2k$	k -component link	No	(none)
$2k + 1$	Torus knot	Yes	Predicted

9 Conclusion

This paper extends the CDFT framework beyond the lepton sector by studying the general $\mathbb{Z}_n$ structure of torus knot families $T(2, n)$.

Theorem 1 proves that $c_n = \sqrt{2}$ for all families, from the maximum-entropy self-consistency condition $\text{Var}(A_k) = \overline{A_k}^2$. This is a universal property of the CDFT vacuum that applies across all particle families.

Theorem 2 proves that $\mathbb{Z}_3$ is the unique family where the Koide invariant $Q_n = 2/n$ is constant on a connected valid domain. For $n \geq 5$, the amplitude $A_k = 1 + \sqrt{2} \cos(\cdot)$ is everywhere negative for at least one mode, so the Koide self-consistency of Paper V does not extend. This explains structurally why the lepton sector is exactly solvable and the cinquefoil sector requires new ingredients.

The Faddeev–Niemi energy hierarchy (4) gives the first-principles energy scales of all torus knot families (Table 1), with $M_5 = 460.4$ MeV the firm prediction for the cinquefoil family.

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A Supplementary Material

Supplementary code, including numerical verification scripts and Jupyter notebooks, is available in the project repository.

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